UPS & battery sizing — UPS-1

REV —

Villa Complex 415V LV Network

Load

Load from	List of loads (below)
Load	**11.1 kVA · 10 kW**
Growth / design loading	20 % / 80 %
Design requirement	16.7 kVA · 15 kW

Loads

Load	Qty	W each	VA each	PF
Servers	10	1000	—	0.9

UPS

Selected UPS	**20 kVA / 18 kW** (output PF 0.9)
Loading today	**56 %** — governed by kVA (55.6 % of 20 kVA · 55.6 % of 18 kW)
Inverter efficiency	94 %

Battery

Type	VRLA lead-acid, end voltage 1.75 V/cell
Backup time	15 min
DC power from the battery	12.77 kW = load kW with growth ÷ 0.94
DC bus	384 V = 32 × 12 V
Calculation basis	estimate
Capacity at this rate	40 % of C10 (typical)
Factors	ageing 1.25 · temperature 1 · design margin 1.1
Required capacity (C10)	28.6 Ah = P × t ÷ 384 V (string) ÷ rate × factors
Battery data issue	None
Selected battery	**32 × 12 V 33 Ah** — 32 blocks, 12.7 kWh
Backup with this battery	**18 min**
Maximum DC current / battery-bus DC breaker	38 A (end of discharge, all strings) / 50 A (≥ 1.25 × I; confirm DC voltage, poles and breaking capacity)

Generic Ah sizing is a preliminary estimate. Entered constant-power tables apply only to the stated battery model, end voltage and reference temperature; runtime is bounded by the supplied rows. Confirm battery and UPS compatibility, temperature correction and charge limits with the manufacturer.